Furthermore, he can use exactly the same rule, Eq. 7.13, to compute the expectation value.

When the density matrix corresponds to a single state, it is a projection operator that projects onto that state. In this case, we say that the state is *pure*. A pure state represents the maximum amount of knowledge that Bob can have of a quantum system. But in the more general case, the density matrix is a mix of several projection operators. We then say that the density matrix represents a *mixed* state.

I have used the term *density matrix*, but strictly speaking, ρ is an operator. It only becomes a matrix when a basis is chosen. Suppose we choose the basis $|a>$. The density matrix is just the matrix representation of ρ with respect to this basis:

$$\rho_{aa'} = \langle a|\rho|a'\rangle.$$

If the matrix representation of $\mathbf{L}$ is $L_{a',a}$ then 7.13 takes the form

$$\langle \mathbf{L} \rangle = \sum_{a,a'} L_{a',a}\rho_{a,a'}. \qquad (7.14)$$

Classical physics also has its notion of pure and mixed states, although they are not called by those names. Just to illustrate, let's consider a system of two particles moving along a line. According to the rules of classical mechanics, we can calculate the orbits of the particles if we know the values of their positions (x_1 and x_2) and momenta (p_1 and p_2) at a certain instant in time. The state of the system is thus specified by four numbers: x_1, x_2, p_1, and p_2. If we know these four numbers, we have as complete a description of the two-particle system as it is possible to have: there is no more to know. We can call this a pure classical state.

Often, however, we don't know the exact state, but only some probabilistic information. That information can be encoded in a probability